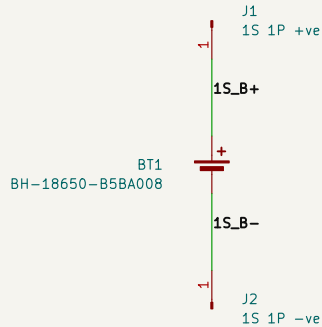


1S 1P MODULE (Single Cell ACIR Baseline)

Baseline 18650 EIS Measurement: $ACIR = R_{cell}$



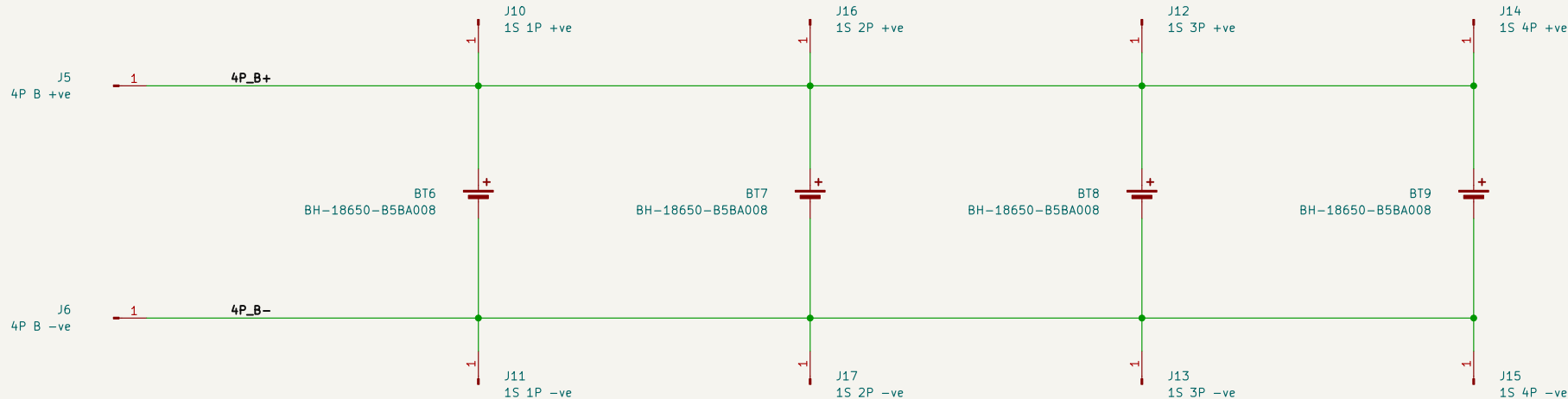
4-Wire Kelvin ACIR Measurement:

Inject 1 kHz AC current across B+ and B- terminals.

Sense AC voltage drop to isolate purely ohmic internal cell resistance.

1S 4P MODULE (Parallel Current Sharing & R/4 ACIR Pack)

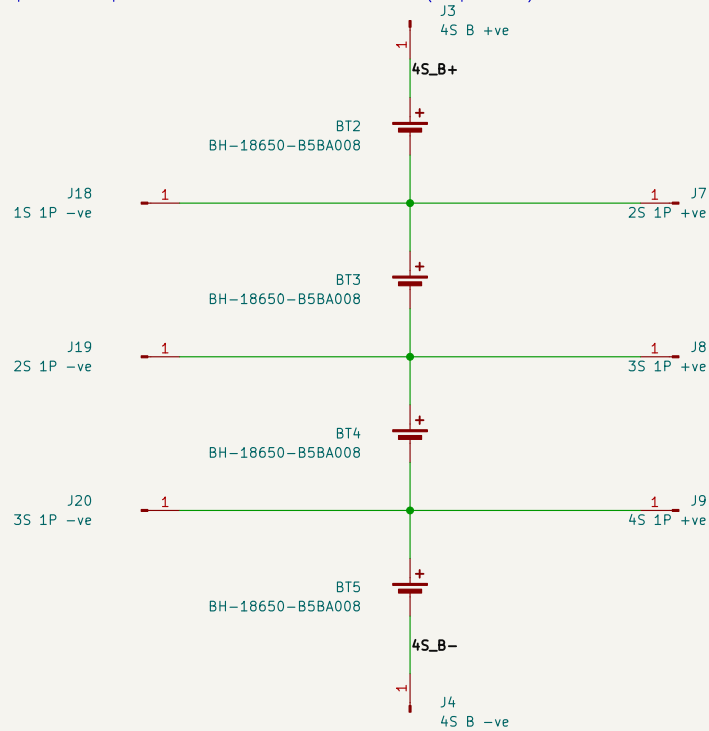
Current Distribution & Branch Kelvin Sensing: $1/R_{eq} = 1/R1 + 1/R2 + 1/R3 + 1/R4$ ($R_{eq} \approx R/4$)



Individual cell branch test points (J10-J17) enable Kelvin characterization of each parallel cell & current sharing.

4S 1P MODULE (Series String – Voltage Stacking & 4x ACIR)

Total Series Impedance: $R_{eq} = R1 + R2 + R3 + R4 + R_{interconnect}$ ($R_{eq} \approx 4 * R$)

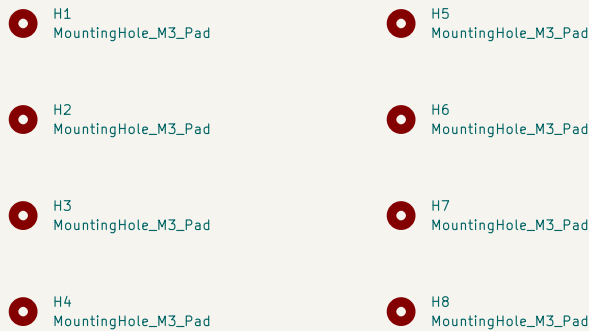


Inter-cell Kelvin probing pairs (J18/J7, J19/J8, J20/J9) allow measuring:

- Individual cell ACIR in-situ under series string current loading.
- Interconnect contact resistance across cell-to-cell busbars.

MECHANICAL MOUNTING HOLES

8x M3 Plated Chassis Grounding / Standoff Fixtures



EIS DESIGN & FABRICATION SPECS

1) Kelvin 4-Wire EIS Architecture:

Separate excitation and sensing paths eliminate test lead resistance.

2) Ultra-Dense Micro-Via Array Stitching:

1,792 stitched vias across high-current copper planes.

Effective parallel capacitance $C_{eff} < 0.001$ pF (negligible).

Halved DC resistance ($R_{dc} / 2$) for maximum measurement SNR.

3) PCB Stackup & Layer Specs:

2-Layer 1.6mm FR4, 1oz Cu, Matte Black Mask, HASL Lead-Free.

Production-Ready Hardware Design & Educational Measurement Kit High-Frequency 1kHz Kelvin 4-Wire EIS Sensing Architecture & Ultra-Dense Via Arrays Configurations: 1S 1P Baseline 1S 4P Parallel Current Sharing 4S 1P Series String Comprehensive 18650 ACIR Characterization Platform Decibels Lab Pvt Ltd		
Sheet: / File: ACIR-Learning-Kit-1-PCB.kicad_sch		
Title: ACIR LEARNING KIT 1 – CHARACTERIZATION & EXPERIMENTATION BOARD		
Size: A3	Date: 2026-09-16	Rev: v2.0
KiCad E.D.A. 10.0.5		Id: 1/1